

Demand and Demand Elasticity

EC306: Intermediate Microeconomics — Day 3

Core reading: Perloff (2020), Ch. 5 and §2.3

Recap and today's payoff

Days 1–2 gave us the chosen bundle as a function of prices and income: the Marshallian demand $x^*(p_x, p_y, m)$.

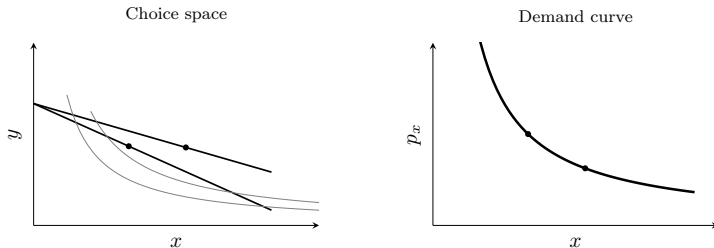
Today we put it to work

- ➊ Trace out the *demand curve* and explain why it slopes down.
- ➋ Decompose any price change into a *substitution* and an *income* effect.
- ➌ Measure responsiveness precisely with *elasticities*, and link them to revenue.

This is the most “applied” day — elasticities are the single most-used number in pricing, tax, and competition work.

From the optimum to the demand curve

Lower p_x , hold p_y and m fixed; the budget line pivots out; the optimum moves; plot x^* against p_x .

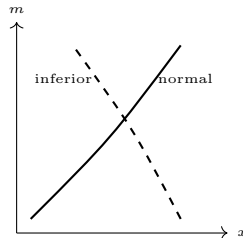


The price-consumption path on the left becomes the demand curve on the right.

Engel curves: how demand responds to income

Holding prices fixed, raise m and plot x^* against m : the **Engel curve**.

- **Normal good:** $\partial x^* / \partial m > 0$. (Most goods.)
- **Inferior good:** $\partial x^* / \partial m < 0$ over some range — demand *falls* as you get richer.



Real inferior goods

Bus travel, instant noodles, supermarket own-brand staples: as income rises, people substitute toward cars, restaurants, and premium brands. Inferiority is about *position in the budget*, not quality.

Decomposing a price change

A fall in p_x does two things at once:

Substitution effect (SE)

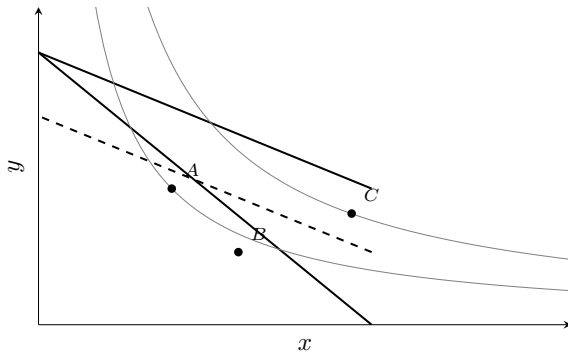
x is now relatively cheaper $\Rightarrow$ swap toward x , *holding utility (or real income) fixed*. Always moves *opposite* to the price: $p_x \downarrow \Rightarrow x \uparrow$. Unambiguous.

Income effect (IE)

A lower p_x raises *purchasing power* $\Rightarrow$ you can afford more of everything. Direction depends on whether x is normal ($\uparrow$) or inferior ($\downarrow$).

Total effect = SE + IE.

The Slutsky picture



$A \rightarrow B$: **substitution effect** (slide along the *old* curve to the new slope). $B \rightarrow C$: **income effect** (jump to the higher curve). $A \rightarrow C$ is the total move.

Giffen goods: when the income effect wins

For a Giffen good the negative income effect (it's strongly inferior) *outweighs* the substitution effect, so demand **rises** when price rises.

The one credible empirical case

Jensen & Miller (2008) found suggestive Giffen behaviour for rice/wheat among very poor households in China: when the staple got dearer, families dropped pricier protein and ate *more* of the staple to hit calorie needs. Requires: a good that is (i) strongly inferior, (ii) a large budget share, (iii) few substitutes. Vanishingly rare — but theoretically real, not a paradox.

Exam trap: “downward-sloping demand” is a *result*, not an axiom. It can fail. Know exactly when.

Own-price elasticity of demand

$$\varepsilon = \frac{\% \Delta Q}{\% \Delta p} = \frac{\partial Q}{\partial p} \cdot \frac{p}{Q}.$$

A *unit-free* measure — the reason we don't just use the slope $\partial Q / \partial p$ (which depends on units).

- $|\varepsilon| > 1$: **elastic** (quantity reacts strongly — luxuries, many substitutes).
- $|\varepsilon| < 1$: **inelastic** (quantity barely moves — necessities, addictive goods).
- $|\varepsilon| = 1$: unit elastic.

Numbers from the real world

Short-run petrol demand ≈ -0.25 (inelastic: you still commute). Cigarettes ≈ -0.4 . Restaurant meals or a specific airline route ≈ -2 to -4 (elastic: easy substitutes). These signs and magnitudes drive tax and pricing policy.

Point vs arc elasticity — and a common error

Point (calculus, one price): $\varepsilon = \frac{dQ}{dp} \frac{p}{Q}$.

Arc (between two observed points, using midpoints):

$$\varepsilon_{\text{arc}} = \frac{(Q_2 - Q_1)/[(Q_1 + Q_2)/2]}{(p_2 - p_1)/[(p_1 + p_2)/2]}.$$

Why the midpoint formula exists

Naive $\% \Delta$ gives different answers depending on which point is the “base”. The midpoint method is symmetric, so a rise from 10→12 and a fall from 12→10 give the same elasticity. Use arc for discrete data, point for a known demand function.

Worked example — linear demand has varying elasticity

$$Q = 100 - 2p.$$

$$\frac{dQ}{dp} = -2, \quad \varepsilon = -2 \cdot \frac{p}{100 - 2p}.$$

p	Q	ε	region
10	80	-0.25	inelastic
25	50	-1.00	unit elastic
40	20	-4.00	elastic

Key lesson

A *straight-line* demand curve does **not** have constant elasticity. Slope is constant (-2); elasticity runs from 0 at the bottom to $-\infty$ at the top, passing through -1 at the midpoint.

Elasticity and total revenue

$TR = p \cdot Q$. Differentiating, $\frac{dTR}{dp} = Q(1 + \varepsilon)$. So:

- Inelastic ($|\varepsilon| < 1$): raise price $\Rightarrow$ revenue **rises**.
- Elastic ($|\varepsilon| > 1$): raise price $\Rightarrow$ revenue **falls**.
- Revenue is maximised at $|\varepsilon| = 1$.

Why governments tax cigarettes and petrol

Inelastic demand means a tax raises lots of revenue with little quantity drop — and the consumer bears most of the burden. The same logic tells a streaming firm whether a price rise grows or shrinks revenue: it hinges entirely on where $|\varepsilon|$ sits relative to 1.

Cross-price and income elasticities

Cross-price: $\varepsilon_{xy} = \frac{\partial Q_x}{\partial p_y} \frac{p_y}{Q_x}$.

- > 0 : **substitutes** (Pepsi vs Coke).
- < 0 : **complements** (printers vs ink).

Income: $\varepsilon_m = \frac{\partial Q}{\partial m} \frac{m}{Q}$.

- > 1 : luxury. $0 < \varepsilon_m < 1$: necessity.
- < 0 : inferior good.

Competition policy uses cross-price elasticity directly

Whether two products are “in the same market” for a merger case is, operationally, a cross-price-elasticity question: if a price rise in A sharply lifts demand for B, they constrain each other.

Your turn (1) — in-lecture

Your turn

Demand is $Q = 240 - 4p$.

- 1 Find the price at which demand is unit elastic.
- 2 A firm currently charges $p = 20$. Should it raise or cut price to increase revenue? Justify with the elasticity.

Your turn (2) — in-lecture

Your turn

For Cobb–Douglas demand $x^* = \frac{a}{a+b} \frac{m}{p_x}$, compute the own-price, cross-price, and income elasticities of x . Comment on each.

What you can now do

- ➊ Build a demand curve and an Engel curve from the optimisation solution.
- ➋ Decompose a price change into substitution and income effects, and state when demand could slope up.
- ➌ Compute and interpret own-price, cross-price, and income elasticities.
- ➍ Link elasticity to total revenue and to real tax/pricing/competition decisions.

Tomorrow

We switch sides of the market. Same optimisation logic — but now the **firm** choosing inputs to make output. Production functions, marginal product, and returns to scale.